Peyman Gholami

Contact Information

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Current Position

PhD Student, Computer Science, UBC — 2021 - present
Vancouver, BC, Canada

Education

University of British Columbia

- PhD., Computer Science — 2021 - present
  - PhD. Thesis: *Enhancing Creative Control in AI generative models*
  - Supervisor: Dr. Leonid Sigal, Dr. Robert Xiao

University of Waterloo

- M.Sc. Vision Science & System Design Engineering — 2016 - 2018
  - M.Sc. Thesis: *Developing algorithms for the analysis of Optical Coherence Tomography images*

Amirkabir University of technology (Tehran Polytechnic)

- B.Sc. Biomedical Engineering (with Honours) — 2011 - 2016
  - B.Sc. Thesis: *Design and Production of an Image Processing Tool for Wound Geometry Measurement and Implementation for Bioprinting*

Research Interests

Artificial Intelligence, Generative Models, Computer Vision, Machine Learning, Natural Language Processing, Large Language Models

Publications

- **P. Gholami**, Robert Xiao, "Streamlining Image Editing with Layered Diffusion Brushes", Proceedings of the IEEE/CVF International Conference on Computer Vision, *ICCV* 2025, 17368-17378
- Xincheng Huang, Dieter Frehlich, Kaseya Xia, **P. Gholami**, Robert Xiao, "GaussianNexus: Room-Scale Real-Time AR/VR Telepresence with Gaussian Splatting, To appear in Proceedings of Annual Symposium on User Interface Software & Technology *UIST* 2025
- **P. Gholami**, Robert Xiao, "Diffusion Brush: A Latent Diffusion Model-based Editing Tool for AI-generated Images", 2023, arXiv:2306.00219.
- **P. Gholami**, Robert Xiao, "AutoDepthNet: High Frame Rate Depth Map Reconstruction using Commodity Depth and RGB Cameras", 2024, arXiv:2305.14731.
- **P. Gholami**, P. Roy, M.K Parthasarathy, V. Lakshminarayanan, "OCTID: Optical Coherence Tomography Image Database", *Computers & Electrical Engineering, 81, 106532*, arXiv:1812.07056, 2020.
- A. Gueddana, **P. Gholami**, V. Lakshminarayanan, "Can A Universal Quantum Cloner Be Used to Design an Experimentally Feasible Near-Deterministic CNOT Gate ?", *Quantum Information Processing 18 (7), 221*, 2019.
- **P. Gholami**, M. Sheikh-hasani, M.K Parthasarathy, J. Zelek, V. Lakshminarayanan, "Classification of Optical Coherence Tomography images for diagnosing different ocular diseases", in *Proc. SPIE BiOS 10483: Multimodal Biomedical Imaging XIII, 1048705*, 2018.

- **P. Gholami**, P. Roy, M.K Parthasarathy, J. Zelek, V. Lakshminarayanan, "Intra-retinal segmentation of Optical Coherence Tomography images using active contours with a dynamic programming initialization and an adaptive weighting strategy", in *Proc. SPIE 10487: Optical Coherence Tomography and Coherence Domain Optical Methods in Biomedicine XXII, 104832M*, 2018.
- P. Roy, **P. Gholami**, M.K Parthasarathy, J. Zelek, V. Lakshminarayanan, "Automated intraretinal layer segmentation of Optical Coherence Tomography images using graph-theoretical methods", in *Proc. SPIE BiOS 10483: Optical Coherence Tomography and Coherence Domain Optical Methods in Biomedicine XXII, 104832U*, 2018.
- **P. Gholami**, M.A. Ahmadi-Pajouh, N. Abolfathi, G. Hamarneh, M. Kayvanrad, "Segmentation and Measurement of Chronic Wounds for Bioprinting", *IEEE Journal of Biomedical and Health Informatics*, 22, no. 4, pp 1269-1277, 2017.
- **P. Gholami** and S. Gorji Kandi, "Color Extraction of Textured Images using Discrete Wavelet Transform", in: *Proceeding of the 6th international congress on color and coating (ICCC2015)*, Tehran, pp. 77-78, 2015.

Honors & Awards

- Several study excellence awards and scholarships including:
  - Four Year Doctoral Fellowship (4YF) Award, (UBC) 2021-2025
  - President's Academic Excellence PhD Award, (UBC) 2022
  - Best Master's Graduate Seminar award (UW School of Optometry & Vision Science) 2018
  - Best undergraduate thesis award (AUT Biomedical Engineering Department) 2016
  - UW International Master's Student Award (UW) 2016 - 2018
  - Science Graduate Student Award (UW) 2016 - 2018
- Ranked top 0.2% among approximately 360,000 participants in Nationwide Iranian University Entrance Exam in the field of Mathematics &Physics 2011
- Ranked 2nd in annual swimming championship of Amirkabir UoT 2014

Work & Research Experience

- **Student Researcher, Google**
  Vancouver, BC, Canada Jan 2026 - present
  - Research Areas: Agentic Skills, Agent loss pattern analysis and benchmarking
- **Student Researcher, Google DeepMind**
  Vancouver, BC, Canada Sep 2025 - Dec 2025
  - Research Areas: RLHF, multi-modal reasoning, data valuation
- **PhD SWE Intern, Meta**
  Menlo Park, CA, USA May 2025 - Aug 2025
  - Research Areas: Multimodal LLMs, Image understanding
  - Enhanced image understanding in Multimodal LLMs using SFT for accurate localization of text and objects in ads images
- **Research Intern, Microsoft Research (MSR)**
  Vancouver, BC, Canada July 2024 - Nov 2024
  - Research Areas: Efficient foundation models, sparse attention
  - Designed and developed a novel adaptive scheme for sparse attention in

transformers
- ▷ Worked on optimizing attention sparsity using dynamic k selection techniques.

⊙ **Graduate Research Assistant: UBC Vision Group, Multimodal User Experience (MUX) Group**
Vancouver, BC, Canada Sep 2020 - present

- ▷ Research Areas: AI generative models, Computer Vision, HCI, Multimodal Learning, NLP
- ▷ Developed a real-time diffison-model based image and video editor, incorporating image editing concepts such as layer masks, visibility toggles, and independent manipulation of layers. (*submitted to CVPR 25*)
- ▷ Developed an interactive 3D Gaussian spatting-based 3D editor using latent diffusion models.
- ▷ Utilizing Representation Finetuning for Creative Control in Large Language Models
- ▷ Created a framework consisting of a hybrid camera setup and an autoencoder for reconstructing low-latency, high-speed depth maps.
- ▷ Designed and implemented a framework to obtain intermediate annotations in sparse Video object segmentation datasets with high frame rate and comparable accuracy.

⊙ **Research Scientist**, **Lattice Semiconductor** (Mirametrix Inc.)
Montreal, QC, Canada Sep 2018 - Sep 2020

- ▷ Principal Investigator of the FaceID, Face Alignment, Eyelids, and Fatigue Metrics components of the Mirametrix Attention Sensing Engine
- ▷ Designed various multi-task neural network architectures and developed analytical solutions for handling large scale datasets
- ▷ Mentored and defined projects for several interns working on deep learning and computer vision, e.g., deep 3D face alignment for automatic labeling of facial landmarks
- ▷ Designed and built a ground truth system for the accurate evaluation of face and head tracking systems

⊙ **Research Assistant: Vision and Image Processing (VIP) Lab**, University of Waterloo
Waterloo, ON, Canada Jan 2017 - Sep 2018

- ▷ Developed novel image processing and ML algorithms for the analysis of various types of medical images
- ▷ Collected and proposed one of the most cited retinal image benchmarks (OCTID) and constructed a platform for training deep neural networks for tracking retinal health
- ▷ Designed a multi-phase classifier for classification of OCT images and fully automated identification of different ocular diseases

⊙ **Graduate Research Assistant: Theoretical & Experimental Epistemology Lab (TEEL)**, University of Waterloo
Waterloo, ON, Canada Sep 2016 - Sep 2018

- ▷ Designed and implemented several visual perception experiments, e.g., relative judgment for autistic subjects, motion coherence, reverse phi phenomenon, etc.
- ▷ Formulated and introduced a novel active contour-based algorithm for fully automated segmentation of retinal layers

⊙ Administrative and financial co-ordinator, *UW Optometry Clinic*

Waterloo, ON, Canada Sep 2017 - Sep 2018

- ▷ Organized optometry clinic schedule by prioritizing workload in the clinic and handling the collaboration between the interns and optometrists, and handling electronic medical records
- ▷ Completed several clinical work requirements, including patient confidentiality policies, Tri-council ethics, and privacy and safety training

⊙ Research Assistant at the Diabetes & Metabolic Diseases Clinic of Endocrinology and Metabolism Research Institute
Tehran, Iran Nov 2015 - Aug 2016

- ▷ Designed and prototyped a visual feedback for controlling bioprinter
- ▷ Proposed a computer vision protocol for wound depth estimation using disparity map

⊙ Research Assistant at Color Control Lab - *Amirkabir University of Tech.*
Tehran, Iran May 2015 - Aug 2016

- ▷ Analyzed the effect of texture on the perceived color of textured
- ▷ Conducted research on color vision, color perception, and color spaces

⊙ Research assistant at the Wound & Ostomy Clinic, *Erfan Hospital of Tehran*
Tehran, Iran Dec 2015 - Jun 2016

Professional Society Activities & Memberships

⊙ Peer reviewer

- ▷ Conference on Neural Information Processing Systems- *NEURIPS*2026
- ▷ IEEE International Conference on Computer Vision - *ICCV*2025
- ▷ IEEE European Conference on Computer Vision - *ECCV*2024
- ▷ IEEE Winter Conference on Applications of Computer Vision - *WACV*2023
- ▷ ACM Symposium on User Interface Software and Technology - *UIST*2023
- ▷ Computer Methods in Biomechanics and Biomedical Engineering: Imaging & Visualization (Taylor & Francis), 2019
- ▷ Journal of Medical Imaging (JMI), 2018
- ▷ Biomedical Optics Express (Optical Society), 2018
- ▷ Heliyon (Elsevier), 2018
- ▷ Burns & Trauma (Springer), 2018

⊙ Reading Group Organizer: MLHCI & HCAI-rg at UBC (2022 - 2023)

⊙ Supervisor of the research team and editor of Tapesh Journal's bioelectric group (the student journal of Biomedical Department of Amirkabir university of Technology) (2014 - 2016)

⊙ Student Volunteer Captain at MobileHCI conference (2022)

⊙ Vice President of UBC's Iranian Student Association (2021)

⊙ Director of Amirkabir University Music House (2012-2013)

⊙ Member of the executive committee in the *19th* & *21th* Iranian Conference on Biomedical Engineering (ICBME 2012 & 2014).

⊙ Member of the Chamber Orchestra of Tehran - violin musician (2002- 2016)

⊙ American Academy of Optometry (AAO) student member

⊙ The International Society for Optical Engineering (SPIE) student member

⊙ Professional Engineers Ontario (PEO) & Ontario Society of Professional Engineers (OSPE) student member

Teaching Experience

- Teaching Assistant
  - Advanced machine learning - CPSC 540, *UBC,* 2021, 2023, 2024, 2025
  - Machine learning - CPSC 340, *UBC,* 2020, 2021, 2022 **(lead TA)**, 2024, 2025
  - Data Structures and Algorithms, 2023
  - Applied Machine Learning - CPSC 330, *UBC,* 2022, 2023 **(lead TA)**
  - Computational Thinking - CPSC 100, *UBC,* 2021, 2022 **(lead TA)**
  - Computer Vision - CPSC 425, *UBC,* 2021
  - Visual Optics, *University of Waterloo,* 2017, 2018
  - Ophthalmic Optics I, *University of Waterloo,* 2016, 2017
  - Principles of Radiology & Radiotherapy, *Amirkabir UoT,* 2015
  - Signal & System Analysis, *Amirkabir UoT,* 2015
  - Computer Programming (C++), *Amirkabir UoT,* 2014
- Workshop Instructor
  - General MATLAB Course
  - Advanced MATLAB - Digital Image Processing
  - LaTeXwriting workshop
- Other experiences
  - Freelance Tutor: Machine Learning, undergraduate CS courses(2016-present)
  - Private Tutoring: teaching mathematics, physics, chemistry and English to high school students (2011 - 2016)
  - Educational Consultant at Cultural Center of Education & Salam High School (2013 - 2015)

Technical Skills

- Programming Languages: Python, C++, JavaScript
- AI Packages, and Libraries:
  - Tensorflow/Keras, PyTorch/Torch, Scikit-learn, Open CV, Pandas, Gradio, Diffusers, TensorRT, WebSocket, transformers
- Cloud computing platforms: Google Cloud, Microsoft Azure, AWS, UBC ARC Sockeye, MAST
- Version controls: GitHub, Perforce, Mercurial
- Operating Systems: Linux, Windows, OS X
- MATLAB: General Matlab, DIP, DSP, Simulink, SISOTOOL, Psychtoolbox
- General Software Packages
  - Microsoft Office, LaTeX, Adobe Creative Suite (Photoshop, Audition, Premier Pro)

Language Proficiency

- English (Native / Bilingual Proficiency.)
- Persian (Native / Bilingual Proficiency.)
- Arabic (Intermediate)

References

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